



Data-Driven Determination of Localized Farmgate Price Volatility in MIMAROPA Condiment Markets

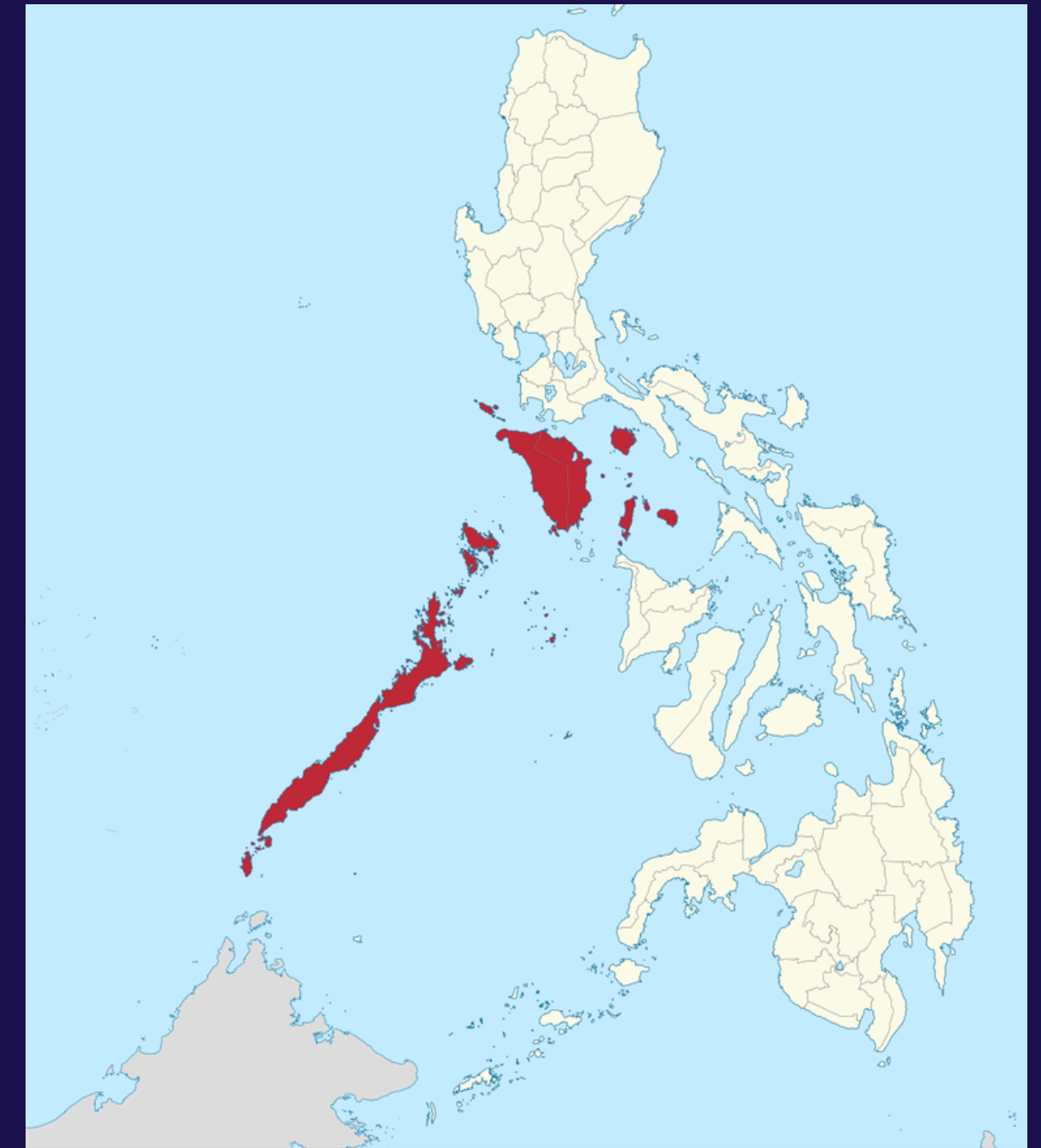
Using Supervised Regression and Spatial Clustering

Presented by: Group 1 | BM2

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The Archipelagic Condiment Economy & Supply Friction

- Agrarian smallholders cultivate high-value condiments (garlic, ginger, and chilis) immediately following the monsoon rice harvest to exploit residual soil moisture.
- Philippine national garlic production (5,830 MT) satisfies only 2.6% of domestic demand, necessitating a 95% import reliance ratio.
- Urban retail consumers purchase garlic at peak valuations of PHP 150/kg, while localized farmgate prices crash to floor valuations of PHP 30/kg.
- Lacking decentralized public cold storage, synchronized seasonal harvests trigger severe localized oversupply gluts and mass crop spoilage.



Scope Boundaries & Operational Data Dictionary

- We focus exclusively on localized household consumer purchasing power and smallholder farmgate revenue.
- Our analysis delimits the scope by explicitly excluding commercial wholesale supply chains, industrial food syndicates, and international export tariffs.
- The operational feature space is structured to define algorithmic ingestion parameters, ensuring all variables (including geolocation, commodity type, and time tensors) are formatted for immediate mathematical execution.

Variable Name	Data Type	Structural & Economic Definition
Geolocation	String	Specific monitoring unit inside the MIMAROPA supply chain.
Commodity	String	Raw condiment crop variety as transcribed in PSA OpenSTAT.
Parent_Class	String	Engineered macro-classification grouping sub-varieties.
Price	Float	Reported monthly farmgate valuation in PHP per kilogram.
Date	Datetime	Standardized, unpivoted monthly timestamp (YYYY-MM-DD).
Month_Index	Integer	Sequential integer offset ($0 \leq t \leq 59$) acting as the continuous time tensor for ML models.

Review of Related Literature

The Conceptual Pillars of the Predictive Pipeline

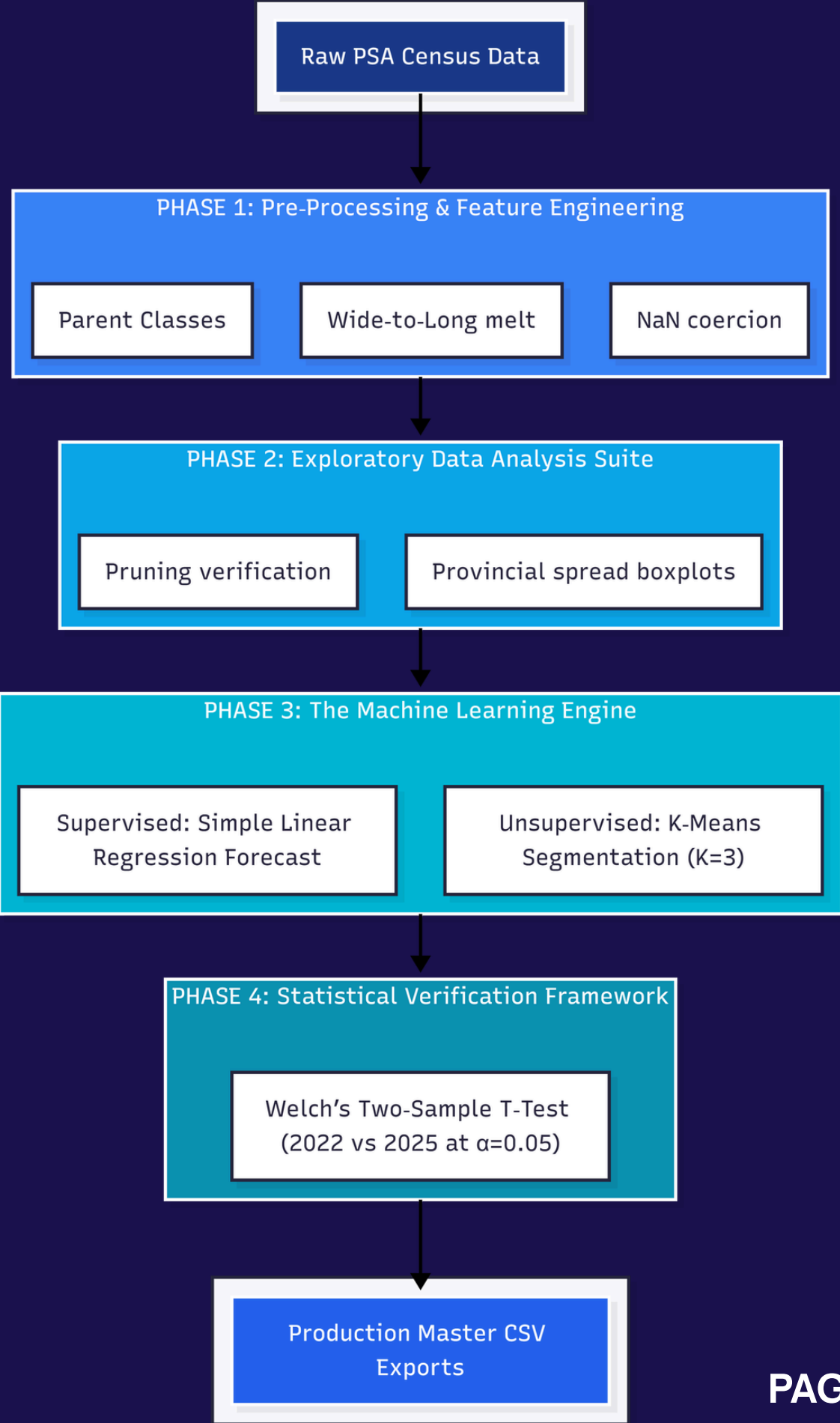
SPATIAL NETWORKS	PRICE DETERMINANTS
Bublik et al. (2026). Optimizing physical trading radius trumps mere dealer density. K-Means audits dealer distance.	Catacutan (2021) & Hinlo & Lee (2026). VAR econometrics isolates gluts. Local holding deficits drive whiplash over import volumes.
ML SUPERIORITY	CORE SYNTHESIS
Theofilou (2025). Machine Learning solves consistency in long-term crop forecasting. ML/DL outperforms linear time-series.	Foundational Premise. Algorithms calculate underlying inflation floors. Diagnostics must pair with cold storage infrastructure.

METHODOLOGY

Computational Methodology Pipeline

We engineered a sequential Python architecture to ensure strict computational reproducibility, guiding the raw census data through four specialized stages of analysis, from initial pre-processing and sparsity auditing to supervised machine learning and final statistical verification.

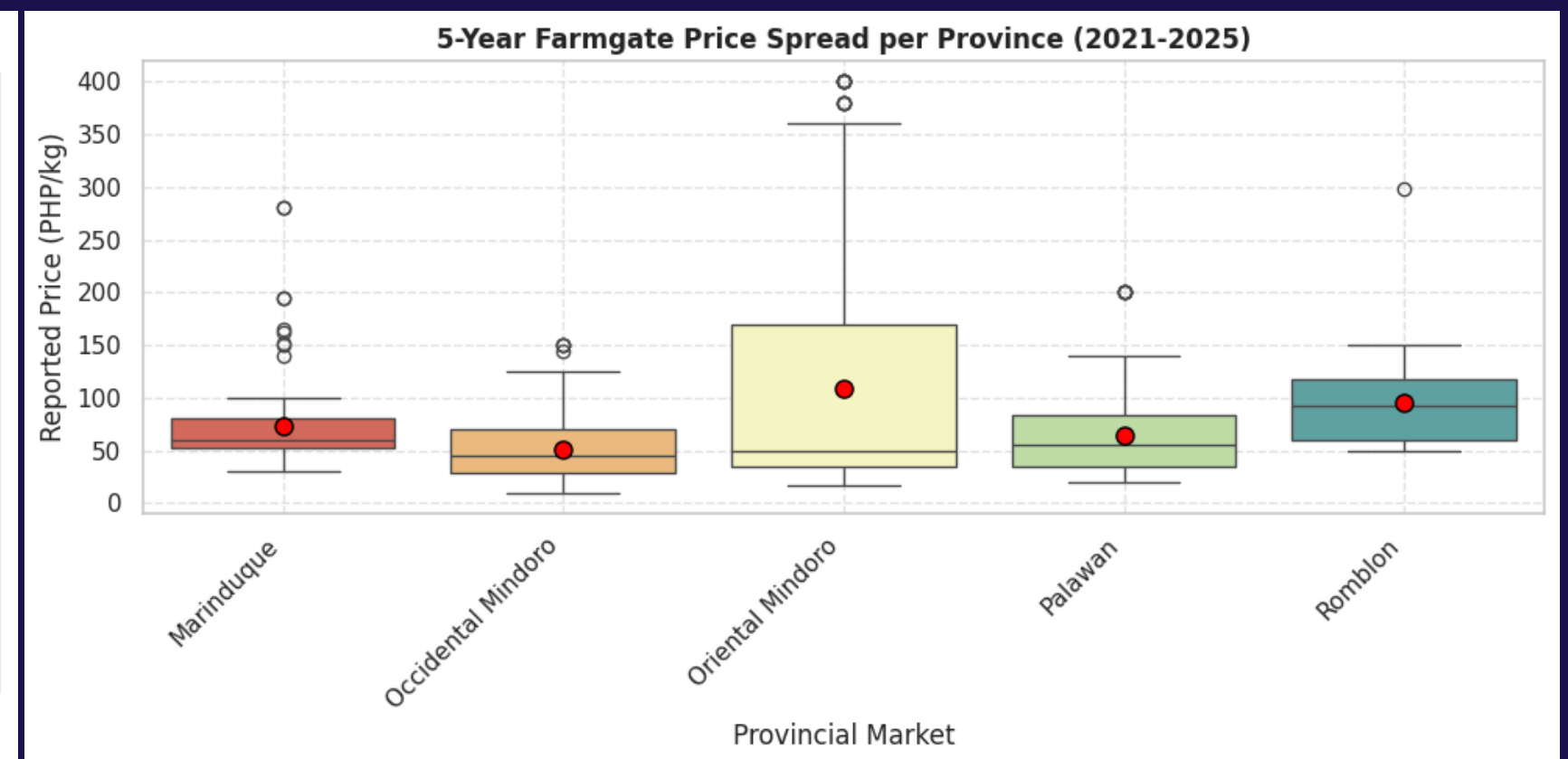
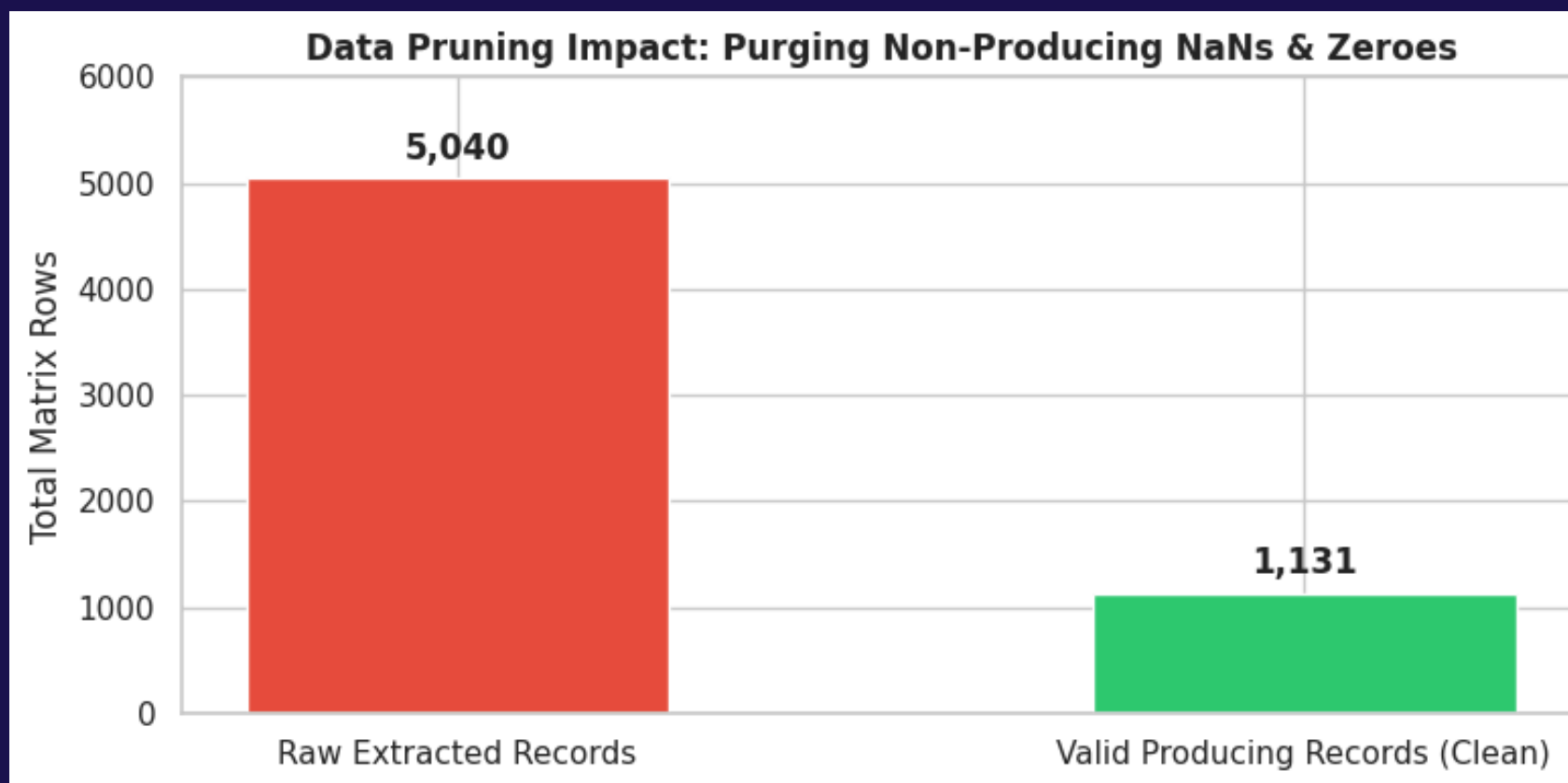
- **Phase 1 (Pre-Processing Engine):** Wide-to-Long dimensional unpivoting, string normalization, and custom Parent_Class aggregation.
- **Phase 2 (Exploratory Data Analysis):** Empirical matrix sparsity auditing and provincial variance distribution mapping.
- **Phase 3 (Machine Learning Core):** Supervised linear forecasting paired with unsupervised K-means spatial segmentation (K=3).
- **Phase 4 (Statistical Verification):** Independent Welch's Two-Sample T-test evaluating historical price drift.



RESULTS AND DISCUSSIONS

Sparsity Audit & Diagnosing Holding Deficits

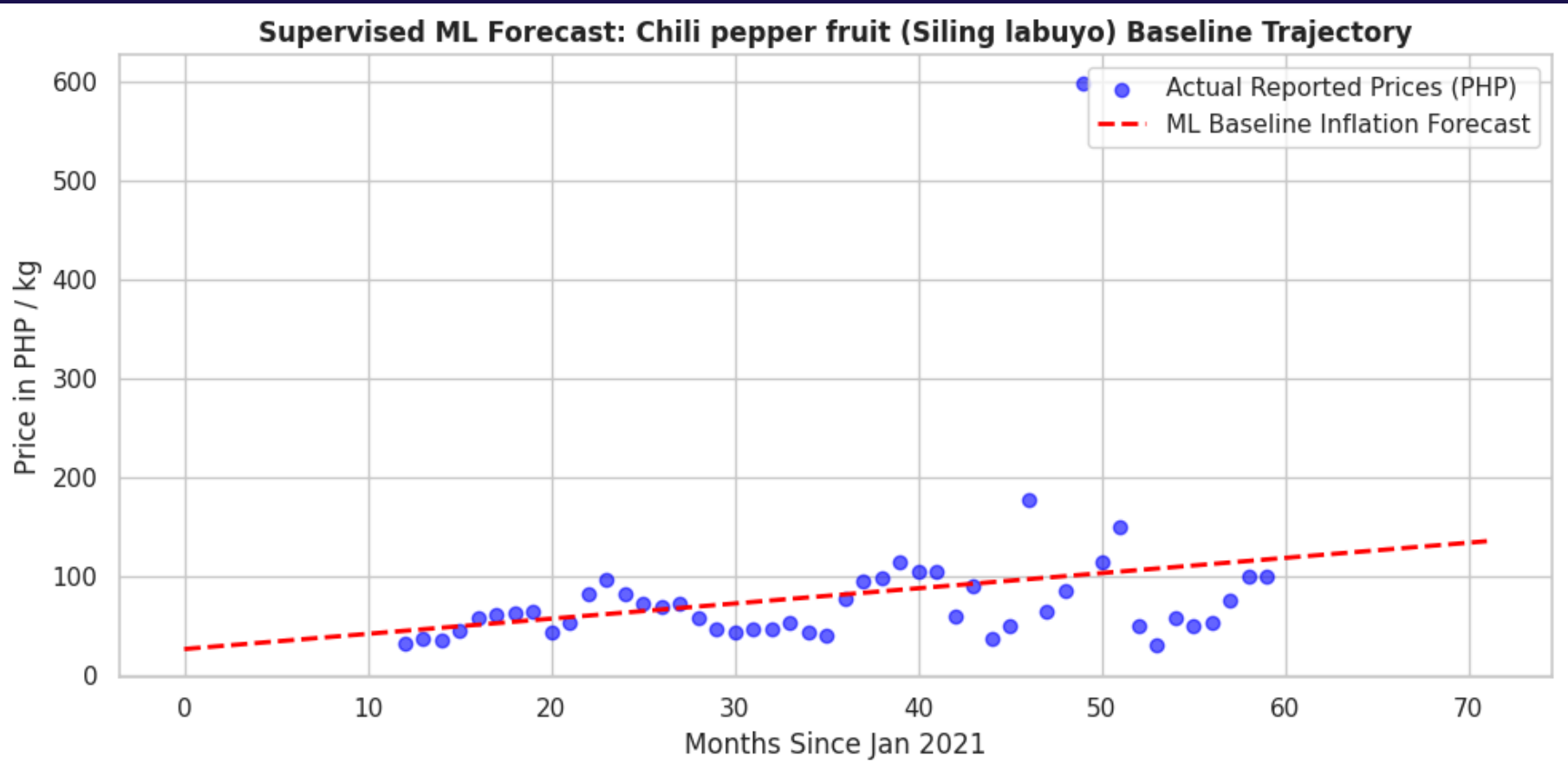
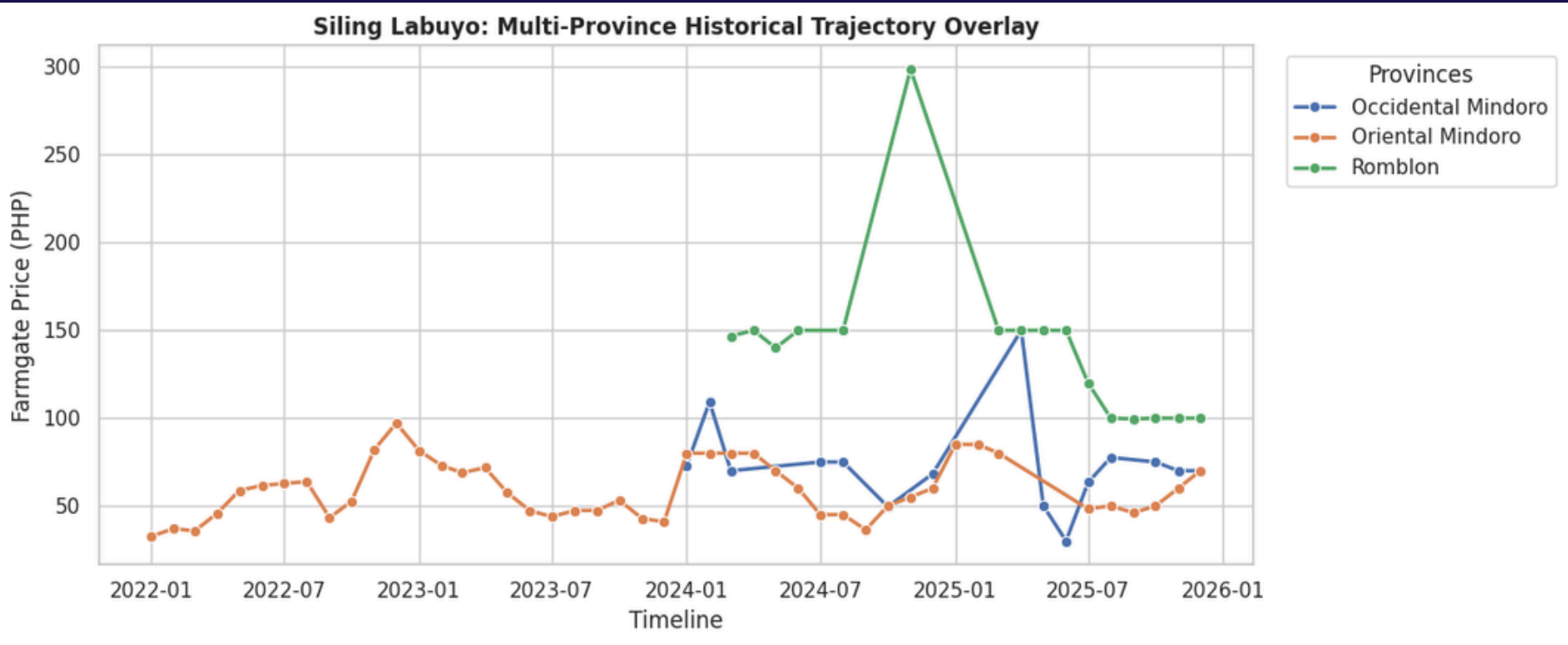
- We condensed the raw census extraction of 5,040 instances into an active production tensor of 1,131 valid records, representing a 22.5% retention rate.
- Purging 3,909 unrecorded zero-production months prevented our estimators from artificially skewing baseline inflation downward.
- The observed boxplot spreads prove severe seasonal price whiplash in Oriental Mindoro and Palawan, which empirically confirms the localized cold storage deficits in those provinces.



RESULTS AND DISCUSSIONS

Trajectory Overlays & Supervised Baseline Forecast

- Our multi-province tracking visualizes that supply shocks are not uniform, as extreme price anomalies, such as the PHP 300 peak near Month 49, are localized to specific hubs.
- While historical averages suggest Romblon is a stable, insulated economy, isolating the specific trajectory of Siling Labuyo reveals extreme vulnerability, with prices spiking to PHP 300.
- Linear regression resolves market noise and proves that the price baseline shifts upward continuously at a rate of +1.53 PHP per month.
- Compounding price floors ensure that the lowest seasonal floor price recorded in 2025 sits permanently higher than the lowest baseline floor recorded in 2022.



RESULTS AND DISCUSSIONS

Spatial Risk Segmentation & Statistical Certainty

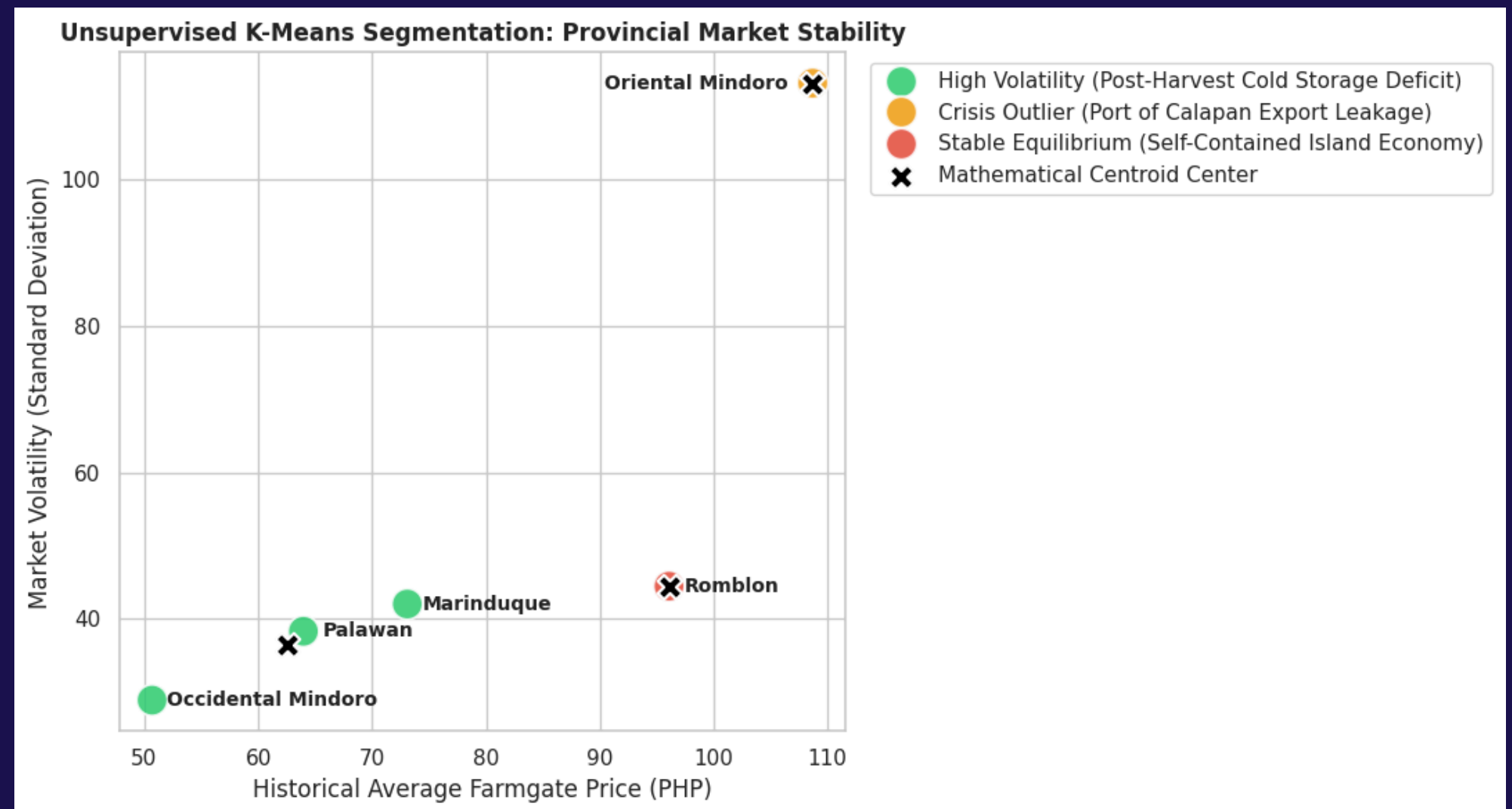
K-MEANS SPATIAL PROFILES

Used **StandardScaler** to convert mean prices (μ) and volatility (σ) into Z-scores.

- **Cluster 0**
 - Romblon (Stable Equilibrium / Insulated).
- **Cluster 1:**
 - Palawan, Marinduque, Occ. Min (High Volatility / Storage Deficit).
- **Cluster 2:**
 - Oriental Mindoro (Crisis Outlier / Calapan Export Leakage).

WELCH'S TWO-SAMPLE T-TEST

- **Sample Cohorts:** 2022 ($n_1 = 24$) vs. 2025 ($n_2 = 39$)
- **Welch t-Stat:** -2.867 (Proves 2025 mean is statistically significantly higher.)
- **Calculated p-Value:** 0.006373

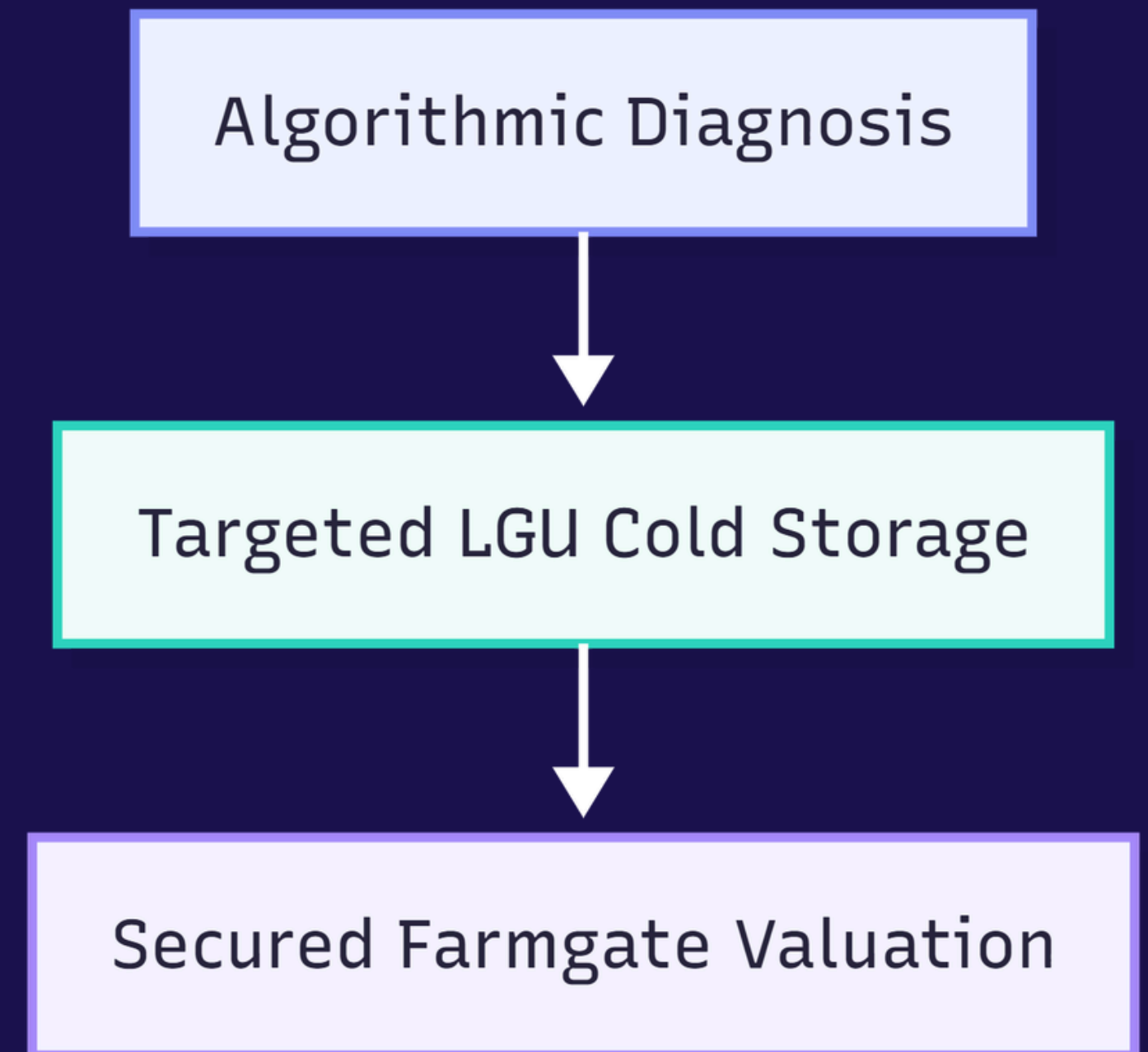


- **Definitive Verdict:** REJECT NULL H_0
- **Certainty Tier:** Proves with 99.36% certainty that regional inflation is a permanent structural shift.

CONCLUSION

Synthesis of Findings & LGU Action Plan

- Baseline inflation is absolute and rising within a 99% statistical confidence tier, while price whiplash is concentrated heavily in maritime transit hubs due to commercial siphoning.
- Increasing raw harvest yield without deploying cold storage infrastructure merely accelerates distressed farmgate dumping.
- Local Government Units must pivot municipal agricultural budgets away from broad fertilizer subsidies and instead deploy decentralized, solar-powered cold storage facilities in Cluster 1 economies.
- Algorithmic intelligence effectively diagnoses the network, but physical infrastructure is required to secure the farmgate valuation.



Research Extension

- We plan to integrate Family Income and Expenditure Survey (FIES) data to evaluate localized consumer purchasing power against farmgate price spikes.
- Future development includes building a multi-dimensional Regional Agrarian Vulnerability Index (RAVI) to rank municipal agricultural risk.
- All four Python execution phases, raw OpenSTAT archives, and sterile master CSVs are packaged natively in an open-access repository for institutional review (**SEE QR**).

THANK YOU FOR LISTENING!



ACCESS OUR PROJECT FILES HERE!